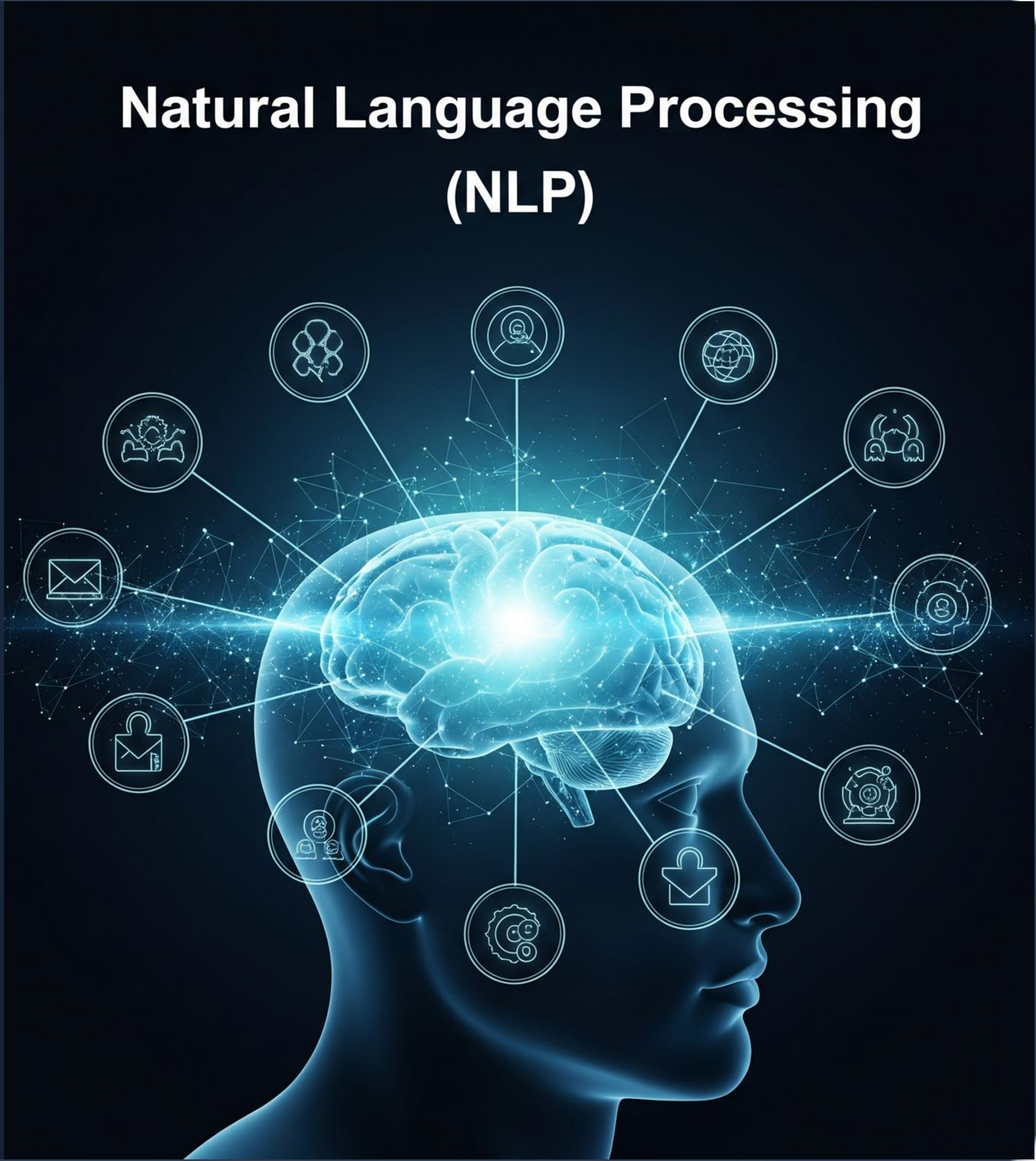
Introduction to Natural Language Processing

Teaching Computers to Speak like Human



Agenda

What is NLP?

Defining the technology and how it bridges humans and machines.

How to Build It

The lifecycle: Cleaning, Vectorization, Training, and Evaluation.

Applications

Chatbots, Translation, and the future with Large Language Models.

What is Natural Language Processing?

Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that gives computers the ability to understand, interpret, and generate human language in the same way human beings can.

Think of it as a **Digital Translator**:

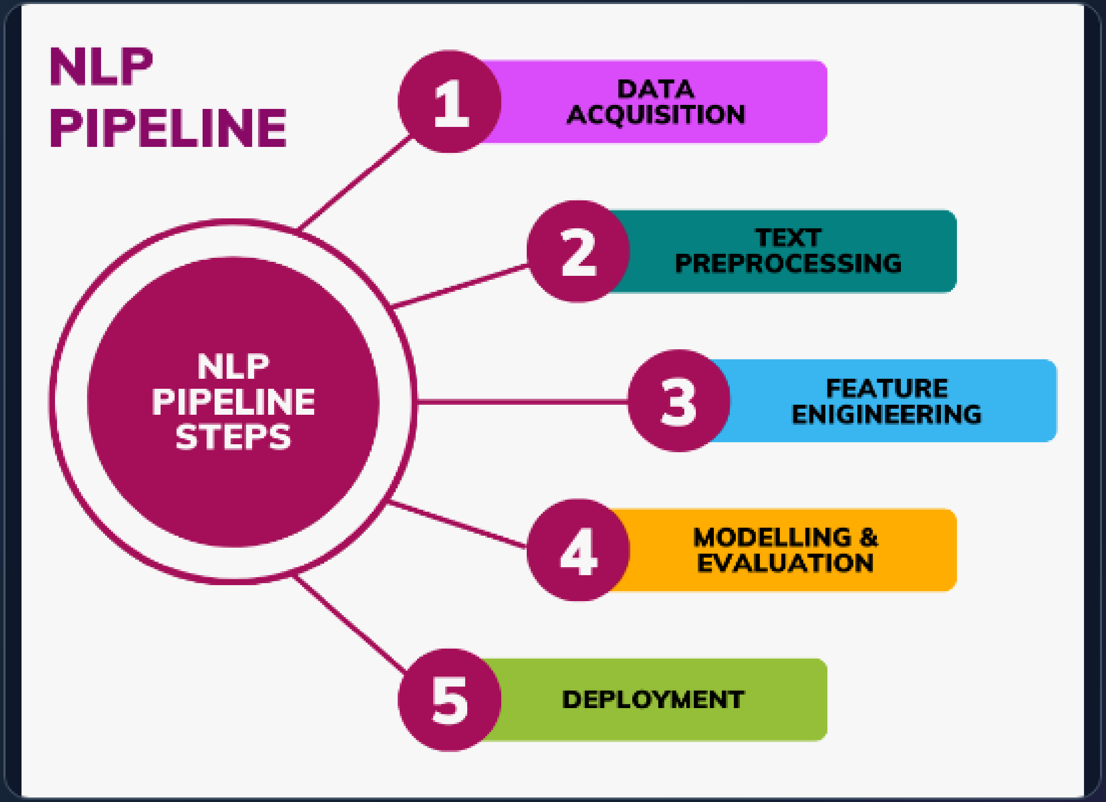
- **Input:** Human Language (Messy, Slang, Ambiguous)
- **Processing:** The "Bridge" (NLP Algorithms)
- **Output:** Computer Understanding (Binary, Structured Data)

The Bridge between Humans & Machines

How Do We Build an NLP Model?

Building an NLP model is like teaching a student for an exam. You need the right books (Data), you need to highlight important notes (Cleaning), and you need to practice (Training).

1 Data Collection → 2 Cleaning → 3 Vectorization → 4 Training

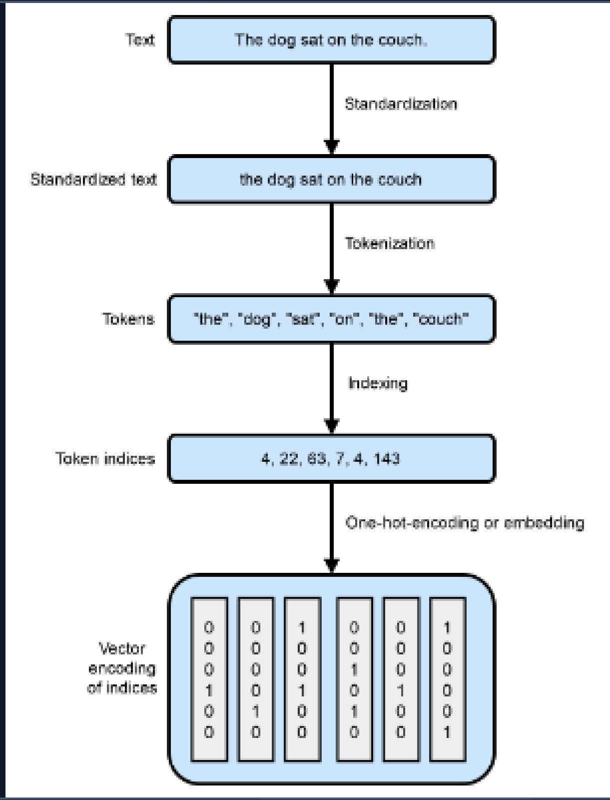


The Basics: Tokenization

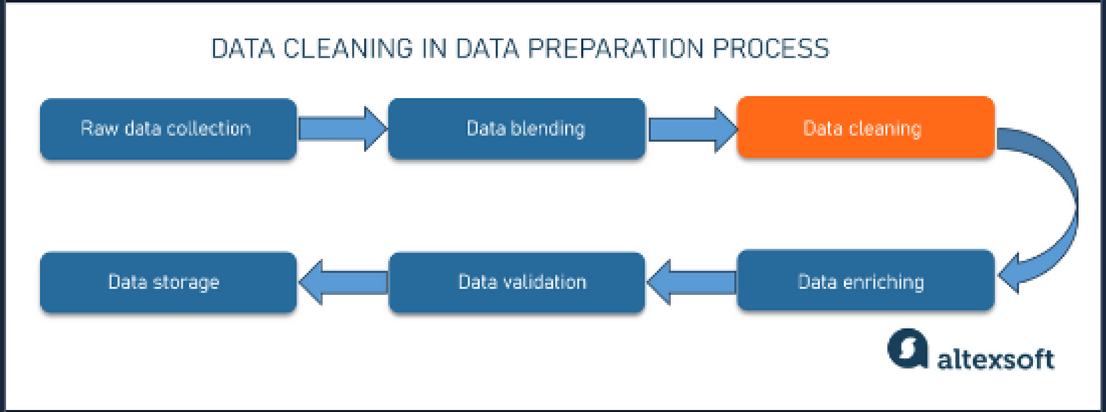
Before doing anything complex, the computer must chop a sentence into pieces called **Tokens**.

"I love AI"
"I" + "love" + "AI"

It also checks **Syntax** (Grammar). Is the sentence structured correctly?



Step 1 & 2: Data Cleaning



Garbage In, Garbage Out

Raw text from the internet is messy. We need to clean it so the model doesn't get confused.

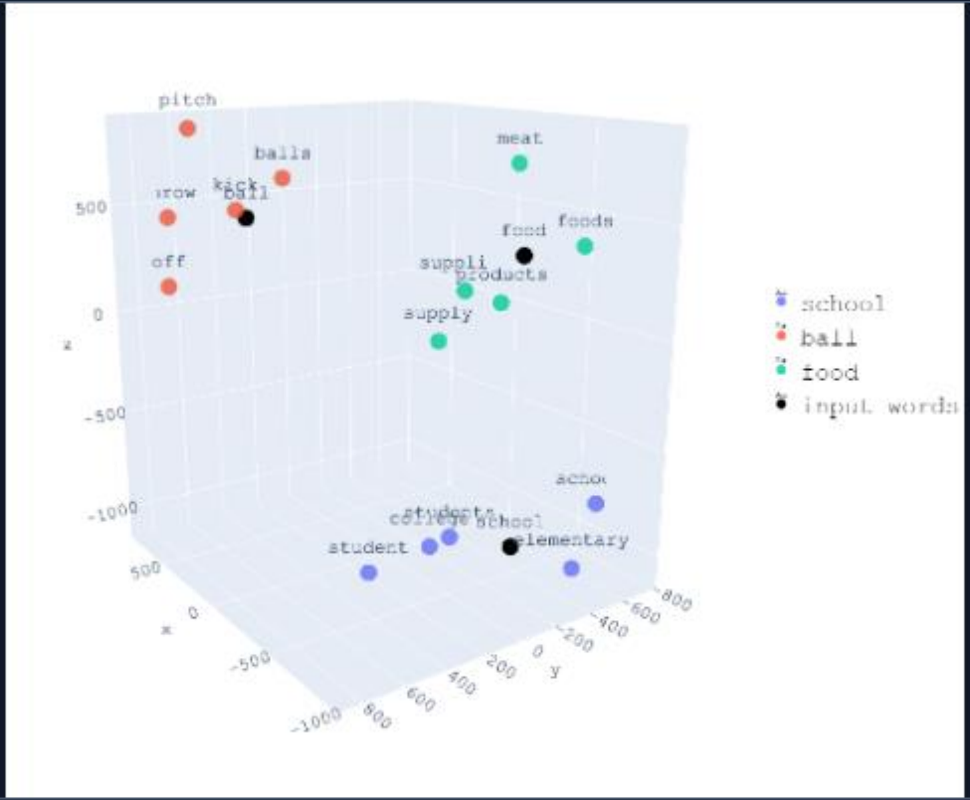
- **Lowercasing:** "Hello" and "hello" should be the same word.
- **Stop Word Removal:** Removing common words like "the", "is", "and" that don't add much meaning.
- **Stemming:** Chopping words to their root.
  "Running", "Runs" -> "Run"

Step 3: Vectorization (Words to Numbers)

Computers cannot read English. They only understand Math.

Vectorization is the process of converting words into lists of numbers (Vectors). Similar words have similar numbers.

King = [0.9, 0.4, 0.7]
Queen = [0.9, 0.4, 0.2]
(Notice they are close mathematically!)



Step 4 & 5: Training & Testing

The Learning Phase

We feed the converted numbers into an algorithm. It looks for patterns (e.g., "words like 'amazing' usually appear in positive reviews").

The Final Exam

After training, we give the model **Unseen Data** (data it hasn't studied) to see if it really learned or just memorized the answers.

Types of NLP Tasks

NLU: Understanding

Natural Language Understanding

The computer comprehends intent. (e.g., Understanding that "I'm freezing" means "Turn up the heat").

NLG: Generation

Natural Language Generation

The computer creates new text. (e.g., ChatGPT writing a poem or a weather report).

NLP in Everyday Life

Voice Assistants

Siri & Alexa use NLP to convert sound waves to text and understand commands.

Predictive Text

Autocorrect and Gmail's "Smart Compose" predict your next word based on probability.

Spam Filters

Email providers scan text to identify "spammy" words and protect your inbox.

Advanced Applications

1. Machine Translation

Converting text from one language to another while keeping the meaning intact (e.g., Google Translate).

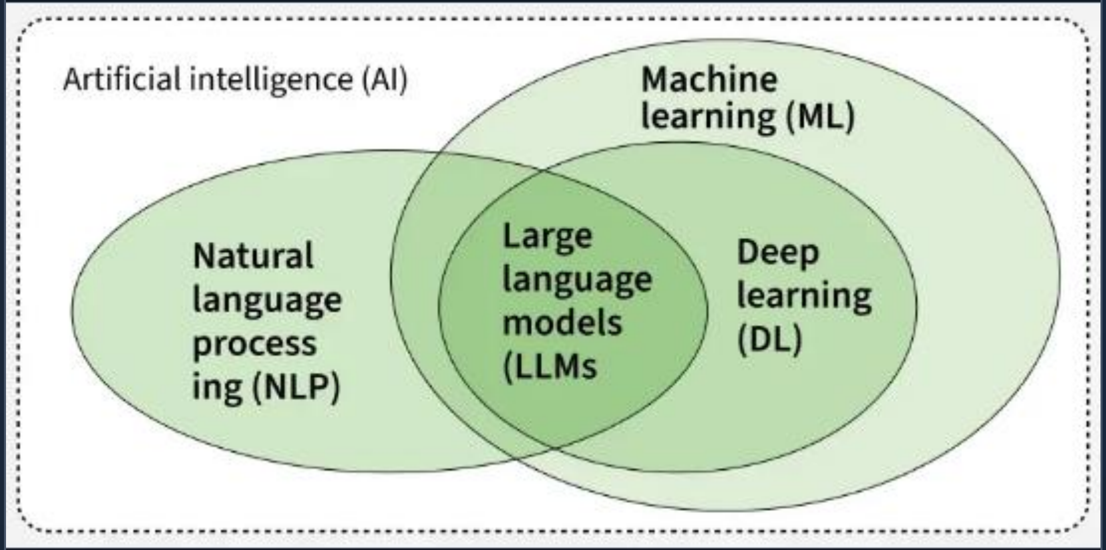
2. Sentiment Analysis

Determining if a piece of text is positive, negative, or neutral. Companies use this to check reviews.

3. Chatbots

Automated customer service agents that can hold a conversation.

NLP vs. LLMs



The Big Picture

Think of **NLP** as the broad field of "Transportation" (Cars, Bikes, Planes).

LLMs (Large Language Models), like ChatGPT, are like a specific **Ferrari**.

- **NLP:** The broad science of language + computers.
- **LLMs:** A powerful tool *inside* NLP that uses massive amounts of data to predict the next word.

Challenges in NLP

1. Ambiguity

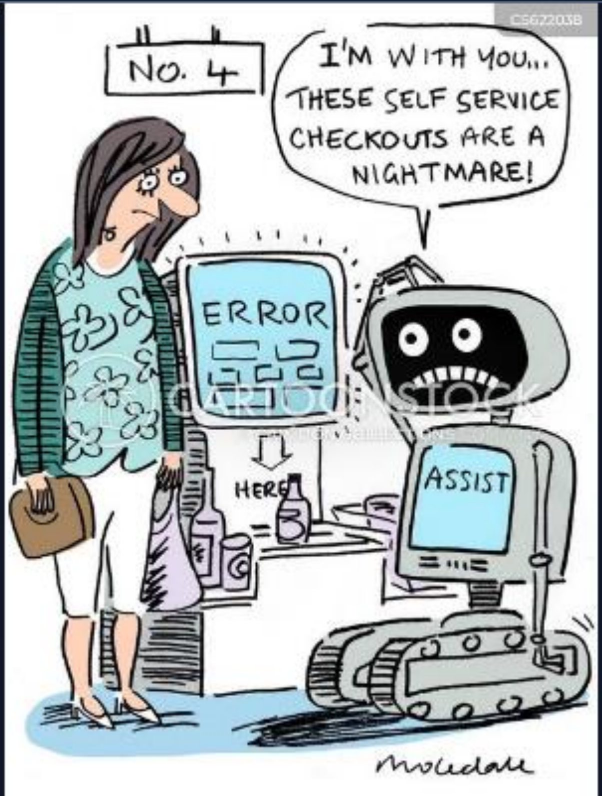
"The bat flew away" - Baseball or Animal? Computers struggle with context.

2. Sarcasm & Irony

They struggle to detect jokes.

3. Slang & Idioms

"Break a leg" means "Good Luck" to us, but a computer might think you want to hurt someone!



The Future of NLP

No Language Barriers

Real-time, perfect translation via earbuds will make language barriers disappear.

Emotionally Intelligent AI

Computers that can understand your mood and respond with empathy, not just logic.